[Let's Titanic!](#) **by Gusty Chickens**
P04: Makers Makin' It, Act II -- The Seequel
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TARGET SHIP DATE: 2026-04-20

PROJECT DESCRIPTION: This game is a Titanic simulation game where your main goal is to save as many Titanic passengers as possible as the captain of the Titanic. The game limits your actions on the ship to 100 moves, which includes moving passengers around the ship, inspecting the passengers' survival rates, etc. forcing you to take risky guesses and prioritize certain actions. Your first inspection for every room is free, so use it wisely! You can load and save your progress, or play a new game as much as you want. Once you are out of moves, you will be given your survival rate and its related metrics.

PROGRAM COMPONENTS

Backend Programming

- **sqlite3 & relevant data tables**: data storage and retrieval
- **__init__.py**: Flask app used for routing, authentication, and accessing databases

Front-end Programming

- **startscreen.html**: landing screen when you first go to the site
- **register.html**: registration page
- **login.html**: login page
- **game_save.html**: displays the saved games for the current user
- **save.html**: will save the current game and return to startscreen
- **lore.html**: plays in a new game, introduces the player's background; will automatically play if you start w/o logging in
- **map.html**: loads the saved game on a map of the Titanic, with number of passengers in the room / max passengers in each room; room capacities will have different classes; passengers start with the room they paid for with their ticket
- **room.html**: loads the opened room from the map with the passenger(s) in that room; players can see their survival rates
- **result.html**: result page with data visualization metrics that appears when you run out of moves; after logging out, will redirect to startscreen

DATABASE ORGANIZATION

DefaultRooms		
INTEGER	id	PK
TEXT	name	
INTEGER	capacity	

DefaultPassengers		
INTEGER	id	PK
INTEGER	survived	
INTEGER	class	
TEXT	name	
TEXT	sex	
INTEGER	age	
REAL	fare	
INTEGER	isAlone	
INTEGER	port	

Users		
TEXT	username	PK
TEXT	password	

Games		
TEXT	username	FK REF Users(username)
INTEGER	id	PK
INTEGER	currLocation	FK REF DefaultRooms(id)
INTEGER	moves	
INTEGER	active	

Passengers		
INTEGER	game	FK REF Games(id)
INTEGER	id	FK REF DefaultPassengers(id)
INTEGER	room	FK REF DefaultRooms(id)

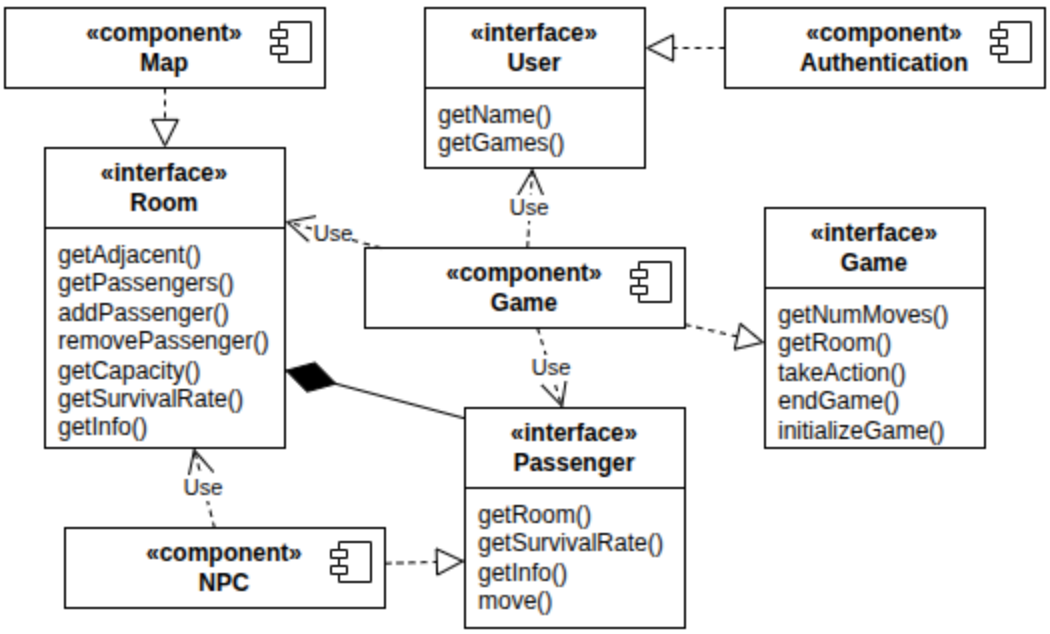
COMPONENT MAP

Map: Moves NPCs and players between rooms

Authentication: Signs users in

Game: Handles actions, end game, and npc / map initialization

NPC: Provide information on NPCs



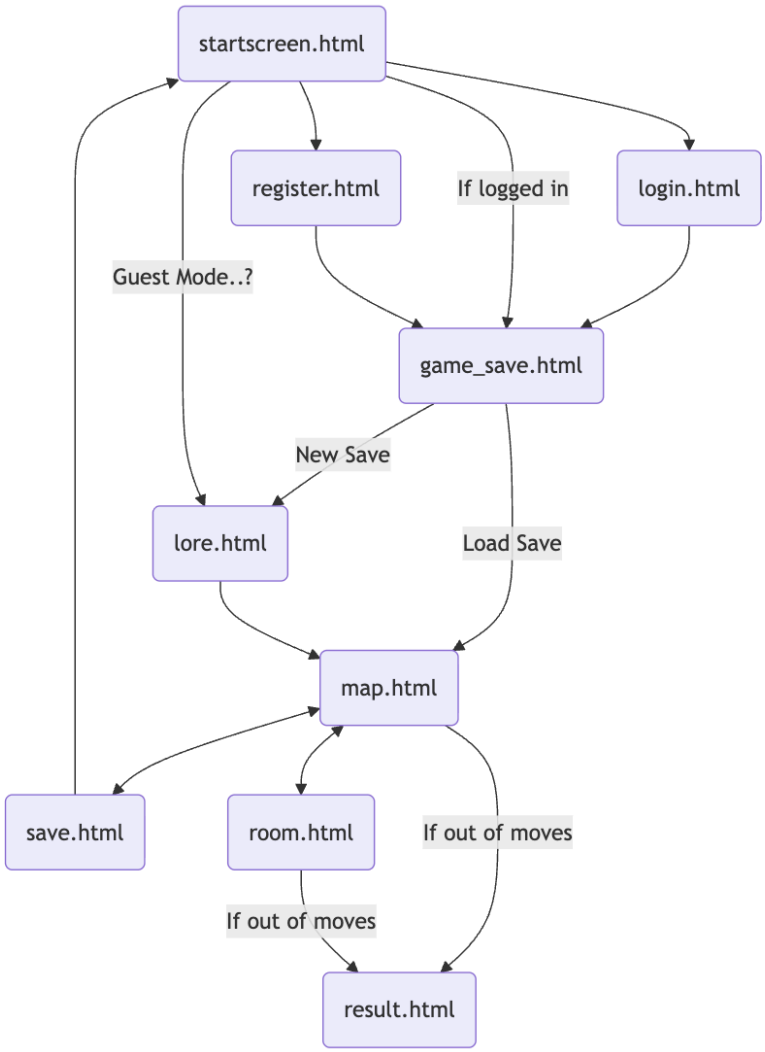
APIs

- We will not be using any APIs.

Extra Mechanics

- Original probability of convincing a crewmate that the ship's about to crash is 8% → increases their survival rate
    - Each passenger convinced raises probability by 1%
- Visiting different rooms will provide you with different items and/or results
- In order to visit a private room / locked room, players need to spend a move to break the lock when entering a room (crew room, passenger room, etc.) if they have the hammer item, or they can use their charisma stat to convince the passenger to let them in

SITE MAP

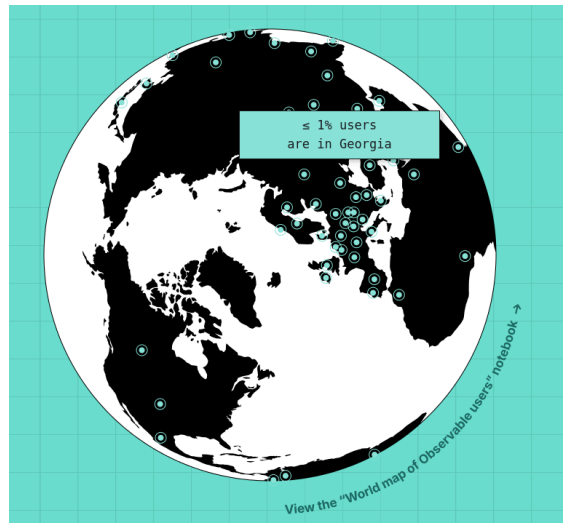


TASK ASSIGNMENTS

Name	**Task**
Emily Mai (PM)	CSS, Register/login, Save page
Carrie Ko	Flask backend, Result page
James Lei	Sqlite db, Lore page
Amy Shrestha	Map page, Rooms page

FRONT-END FRAMEWORK

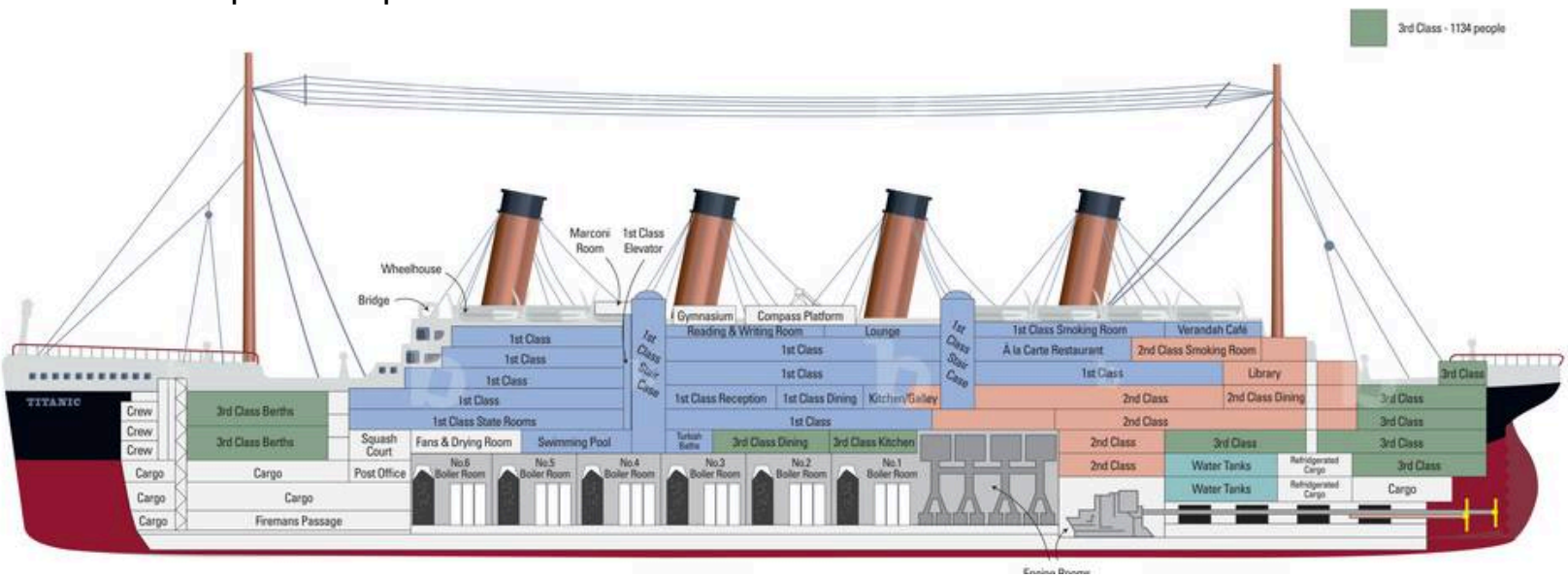
- **Bootstrap** (if time permitting)
- **chart.js**
    - Visualize survival data in animated line charts of
        - player v. real world
        - player v. other players
- (if time permitting) **d3.js**
    - Visualize an interactive map of players (users can choose their region in startscreen) -> example shown on the right



DATA VISUALIZATION LIBRARIES

- **Titanic Survival Datasets**: used to get Titanic NPC passengers (will utilize age, pclass, sex, passengerid, survived, isAlone, and fare columns)
    - Titanic Survival Data: https://www.kaggle.com/datasets/omarmagdy711/titanic-survival-data
    - Titanic Survival Prediction Dataset: https://www.kaggle.com/datasets/aliiihussain/titanic-survival-prediction-dataset
    - Titanic Dataset: https://www.kaggle.com/datasets/brendan45774/test-file
- (if time permitting) **Weather Dataset**: used to get weather for the day and the damage it does to the Titanic every day / hour
- (if time permitting) **Materials / Food Dataset**: used for loot drops, players can use materials to repair the Titanic that's losing HP due to the weather
    - For loot drops:
        - Can find in cargo rooms
        - Can find by getting hammer drop and breaking into passenger's rooms
- (if time permitting) **Pirates Dataset**: used to get pirate NPCs, used in random encounters

Map & Components



Each action takes a move:

- Compass platform (play a gacha to drive the ship)
  - 45% chance nothing happens, 42% chance of losing 2 moves instead of one, 10% chance of increasing survival rate, 2% of instant death, 1% of instant win!!!
- Library
  - Reading books increases charisma by 2% but costs a move
  - And additional 10% chance of finding location of an item
- Gymnasium / swimming pool / squash court
  - 10% chance increase to not consume a move when acting
  - Locked to once per room, doing this consumes an action
- Water tanks & boiler rooms
  - 30% of finding a hammer
- Crew rooms / post office
  - Entering it has an 100% chance of finding the master key, which allows players to go into rooms without spending a move (but requires hammer to break into the room first)
- Kitchen
  - Visiting restores 5 (?) moves and you eat all the food
- Refrigerated cargo
  - Can visit to restore the kitchen (once per game), refills all the food in the kitchen